

4. (a) Limestone quarrying is an important business in the UK. Quarrying creates jobs in areas where there are often limited opportunities. There is a huge demand for the products of quarrying, such as building stone and cement. Good roads and rail links are needed for transporting the products of quarrying. Thousands of people are employed in quarrying and related industries.

Many people argue against the opening of a quarry in their area despite all the benefits of limestone quarrying. Explain **two** reasons local people might use in an attempt to stop the opening of a new quarry in their area. [2]

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- (b) When limestone is heated in a lime kiln thermal decomposition takes place.

Write a balanced **symbol** equation for this reaction. [1]

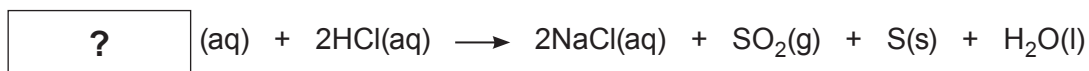
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- (c) Calcium silicide is a compound containing calcium and silicon only. A sample of calcium silicide was found to contain 2.0 g of calcium and 2.8 g of silicon.

Calculate the simplest formula for this compound. You **must** show your working. [3]

Simplest formula

4. (a) Dilute hydrochloric acid reacts with sodium thiosulfate to make the products shown in the equation.



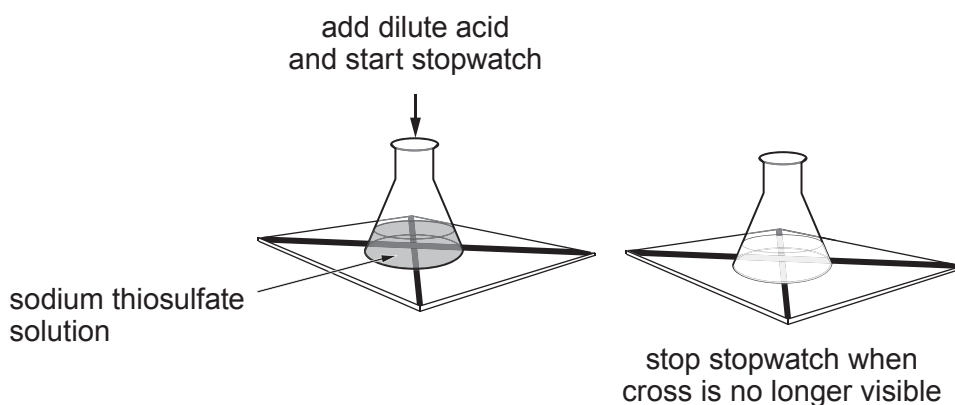
- (i) Use the equation to work out the formula of sodium thiosulfate. [1]

Formula

- (ii) The symbol (aq) in the equation tells us that the substances are aqueous. What is meant by this? [1]

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- (iii) The rate of this reaction can be studied as shown in the diagram.



Use information **from the equation** to explain why the cross disappears. [2]

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(b) A student studied the effect of temperature on the rate of this reaction. He obtained the following results.

Temperature (°C)	Time taken for cross to disappear (s)			
	1	2	3	Mean
15	130	128	129	129
30	53	53	53	53
45	21	29	23	24.3
60	7	7	6	6.7

(i) Another student said that one of the mean values was incorrect. Identify the incorrect mean. Give your reasoning. [2]

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(ii) State what conclusion can be drawn about the effect of temperature on the rate of this reaction. Explain your conclusion using particle theory. [3]

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4. Alfred Wegener proposed the theory of continental drift.

(a) State the **evidence** that Wegener used to support his theory and explain why other scientists refused to accept it. [3]

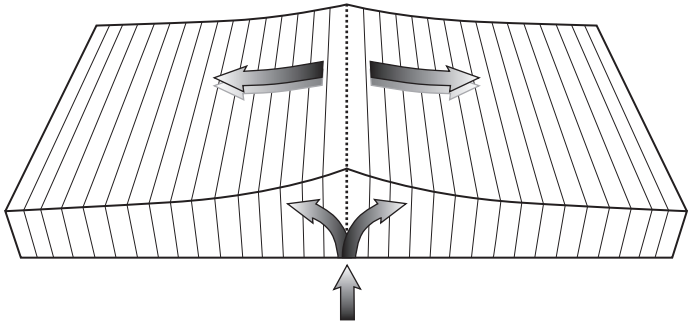
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(b) In the 1960s scientists studied the ocean floor at a constructive plate boundary. Their observations are summarised in the following diagram.



(i) Describe what is happening at this boundary. [2]

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(ii) Label the diagram clearly to show the **oldest** rock. [1]

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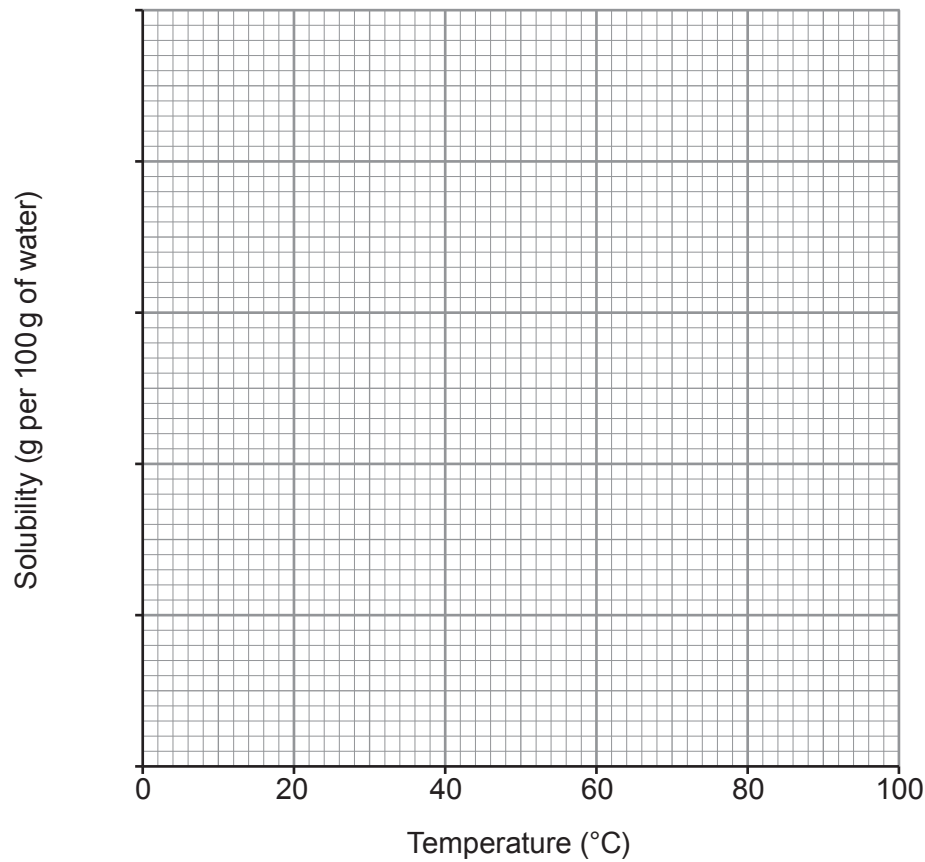
4. (a) The table shows the solubility of potassium nitrate in water at different temperatures.

Temperature (°C)	Solubility (g per 100 g of water)
0	13
20	32
40	64
60	110
80	169
100	246

(i) Choose a suitable scale for the y-axis and plot the data on the grid.

Draw a line of best fit.

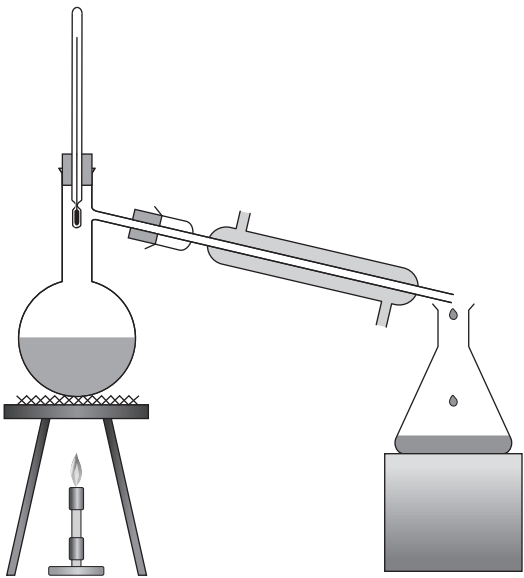
[4]



- (ii) Calculate the mass in grams of potassium nitrate that would be crystallised if 250 cm³ of a saturated solution were cooled from 80 °C to 30 °C. [3]

Mass = g

- (b) Ethanol can be separated from water by distillation using the apparatus shown.



Explain how ethanol and water are separated by distillation. [2]

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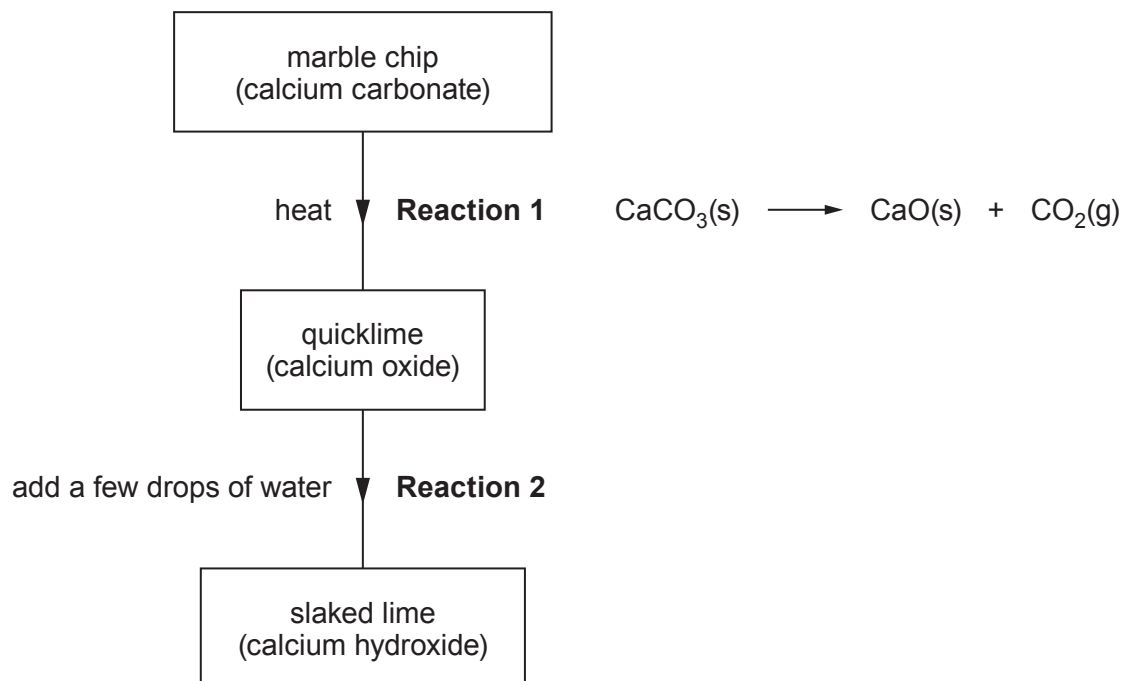
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4. (a) The flow diagram shows the reactions necessary to make slaked lime (calcium hydroxide) from a marble chip (calcium carbonate).



- (i) In **Reaction 1**, one compound breaks down when heated to form two products.

Complete the name of this type of reaction.

[1]

thermal

- (ii) Give **two** observations you would expect to make as **Reaction 2** happens.

[2]

1

2

- (iii) Complete the symbol equation for the reaction between calcium oxide and water in **Reaction 2**.

[1]

..... + H_2O $\longrightarrow$



Examiner
only

(b) (i) The relative formula mass of calcium hydroxide is 74.

Calculate the number of moles of calcium hydroxide in 2.96 g. [2]

Number of moles = mol

(ii) Explain why calcium hydroxide is used to treat some soils. [2]

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